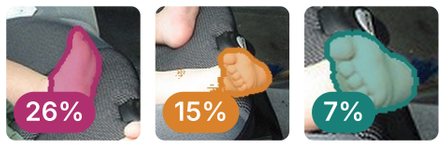
(a) Part grounding

Q *The left foot of the person.*



Top-3 candidates (share)



Box + SAM 3 IoU 0.00

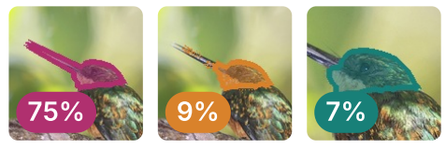


Ours IoU 0.90 dashed: ground truth

Q *The head of the bird.*



Top-3 candidates (share)



Box + SAM 3 IoU 0.24



Ours IoU 0.87 dashed: ground truth

(b) Reasoning

Q *If I want to pick up the knife, which part in the picture can be used?*



Box + SAM 3 IoU 0.22



Sa2VA-4B IoU 0.21



Ours IoU 0.97 dashed: ground truth

Q *We cannot breathe underwater, so diving requires additional equipment to help people breathe while underwater. What in the picture can help human accomplish this task?*



Box + SAM 3 IoU 0.00



Sa2VA-4B IoU 0.05



Ours IoU 0.94 dashed: ground truth

(c) Editing

Q *Craft the ram with fine ceramic.*



Our mask **Edit with our mask** **Edit with Box + SAM 3 mask**

mask IoU: ours 0.98, Box + SAM 3 0.00

Q *Change the building's exterior color to a soft beige.*



Our mask **Edit with our mask** **Edit with Box + SAM 3 mask**

mask IoU: ours 0.99, Box + SAM 3 0.00